

Using High-Speed Nasopharyngoscopy to Quantify the Bubbling Above the Velopharyngeal Valve in Cases of Nasal Rustle

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Abstract

Objective: The loud and severely distorting form of audible nasal emission (commonly known as nasal turbulence or nasal rustle) typically occurs with a small velopharyngeal opening during production of pressure-sensitive consonants. The purpose of this study was to determine whether bubbling of the secretions, which commonly occurs on the superior aspect of the velopharyngeal port when there is a small opening, is a periodic process that can generate sound in the nasal cavity.

Participants: Ten pediatric patients were included in the study. All participants had normal articulation and resonance but exhibited audible nasal emission characterized as nasal rustle.

Measures: For each participant, high-speed video (HSV) nasopharyngoscopy and acoustic signals were recorded simultaneously. The acoustic recordings were captured in a manner similar to nasometry using nasal and oral microphones connected to a separation plate. Spectral analysis of the audio recordings and the HSV images was used to determine correlation between the acoustic and visual measurements.

Results: This study showed that secretion bubbling is a periodic process and its frequency, measured from the HSV data, was also captured by the acoustic measurements. The nasal acoustic signal correlated more strongly with the video of bubbling than the oral acoustic signal in the majority of the cases where bubbling occurred.

Conclusion: These findings are strong evidence that secretion bubbling plays a significant role in the mechanism that generates undesired sound in the nasal cavity. Further work is needed to determine whether this sound is perceived as nasal rustle.

Keywords

high-speed videoendoscopy, spectral characteristics, nasal turbulence, nasal rustle, mucus secretion displacement

Introduction

During normal speech production, the velopharyngeal (VP) valve closes completely on oral speech sounds so that both sound energy and airflow are directed entirely into the oral cavity. If complete VP closure is not achieved due to velopharyngeal insufficiency (VPI), there will be a leak of the airstream into the nasal cavity (eg, nasal emission). If the opening is fairly large, there will also be excessive sound in the nasal cavity, resulting in hypernasality. Nasal emission is inaudible with a large VP opening and barely audible with a medium-sized opening. With a small VP opening, however, nasal emission is not only audible—it can also be very loud and distracting (Kummer et al., 1992; Kummer et al., 2003).

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This severe form of nasal emission was called “nasal turbulence” by McWilliams (1982) and “nasal rustle” by Mason and Grandstaff (1971). Regardless of the term used, most authors agree that the most severe form of nasal emission is associated with a small VP opening during oral consonant production (Kummer et al., 1992; Kummer et al., 2003; Peterson-Falzone et al., 2006; Zajac and Preisser, 2016). What is up for debate, however, is the actual mechanism that produces this undesired sound in the nasal cavity.

McWilliams (1982) suggested that severe audible nasal emission (nasal turbulence) is caused by marked resistance to the airstream where nasal resistance was defined as occurring “when airstream is greater than the space through which it must circulate.” McWilliams speculated that the fast-moving airflow in the posterior part of the nasal cavity produces this severe form of nasal emission.

Trost (1981) observed blurring of the VP port in radiographic images from participants with repaired cleft palate who were diagnosed with compensatory misarticulations for oral fricatives. She hypothesized that the blurring could be due to velar vibrations (flutter) and later noted that a similar sound can occur due to VPI (Peterson-Falzone et al., 2006). The term “velar flutter” suggests that the airflow through the small VP openings can excite the margins of the velar tissue to vibrate, thus producing sound by a mechanism similar to vocal fold vibrations. Zajac and Preisser (2016) argued that the periodicity in their audio recordings (shown as vertical striations in the spectrograms) occurred due to velar flutter. However, their study did not include any direct measurements (eg, using nasopharyngoscopy imaging to show the velar tissue vibrating) to support their claim. In addition, it should be noted that periodicity in signals from an indirect measure of velar function, such as an audio recording, can be generated by other physical mechanisms (eg, bubbling).

Finally, Kummer and colleagues noted that bubbling of secretions on the superior aspect of the VP port often occurs with small VP openings. Therefore, they suggested that cases of severe audible nasal emission, which they called nasal rustle, are produced by the interaction between the airflow and the mucus secretions in the nasopharynx (Kummer et al., 1992; Kummer et al., 2003). However, their study utilized perceptual evaluation and did not include acoustic or physiologic measures that could show that such interaction exists.

Given the uncertainty about the physical mechanism responsible for producing the severe and undesired sound in the nasal cavity in cases of a small opening in the VP port, the aim of this investigation was to study and further define this dynamic process. In this study, we investigated the secretion bubbling that occurs in the nasopharynx when the airflow is released through a small VP opening. We hypothesize that the secretion bubbling is a periodic process that can generate sound in the posterior nasal cavity. To test this hypothesis, multimodal data from simultaneous measurements of acoustics and high-speed nasopharyngoscopy imaging were collected and analyzed.

Methods

This study protocol was approved by the institutional review board of Cincinnati Children’s Hospital Medical Center.

Participants

Participants for this study were recruited from patients seen in the VPI Clinic at Cincinnati Children’s Hospital Medical Center. All patients received a speech/resonance evaluation by a speech-language pathologist who specializes in cleft palate, resonance disorders, and VPI. Criteria for selection of participants for this study included correct articulation placement, normal resonance, and an obligatory nasal rustle during the production of all pressure-sensitive consonants (ie, plosives, fricatives, and affricates). Patients with phoneme-specific nasal emission due to the production of a posterior nasal fricative were not included in this study. Ten participants met the criteria, including 5 boys and 5 girls, ranging in age from 5 to 10 years old with a mean of 6.7 years. Because this clinic is for children with noncleft VPI, none of the patients had a history of overt cleft palate. However, 4 had an unrepaired submucous cleft, 3 had VPI for unknown reasons, 1 had 22Q11.2 deletion syndrome, 1 was postadenoidectomy, and 1 had a history of postnatal stroke. Of the 10 patients, 5 had persistent VPI despite a pharyngeal flap and 1 had persistent VPI despite a pharyngeal injection.

As part of the routine clinic visit, a standard nasopharyngoscopy evaluation was done jointly by an otolaryngologist and a speech-language pathologist, and the relative size of the opening was rated. In all recruited participants, both clinicians rated the VP opening during speech as small, with visible bubbling of secretions on pressure-sensitive sounds. A standard evaluation for the opening size, as proposed by (Golding-Kushner et al., 1990), could not be carried out because the location of the velum at rest was outside the image captured with nasopharyngoscopy.

Procedures

Simultaneous high-speed video nasopharyngoscopy and audio measurements. For each participant, both high-speed video (HSV) nasopharyngoscopy and audio measurements were conducted to collect video images and acoustic data (Figure 1). These data were collected simultaneously so that the acoustic data that exhibited incidents of nasal rustle could be correlated with visual information from the high-speed camera.

For the HSV nasopharyngoscopy, a flexible scope (outer diameter: 2.2 or 3.6 mm) was connected to an HSV camera (MIRO 310; Vision Research, Inc., Wayne, New Jersey), and the nasopharyngoscopy images were captured at a rate of 5000 frames/second using 304 × 256 pixel resolution. This rate is significantly faster than a standard nasopharyngoscopy procedure, where images are typically captured at rates of only 30 to 60 frames/second. Example videos from the HSV data are attached to this article so that the reader can appreciate the

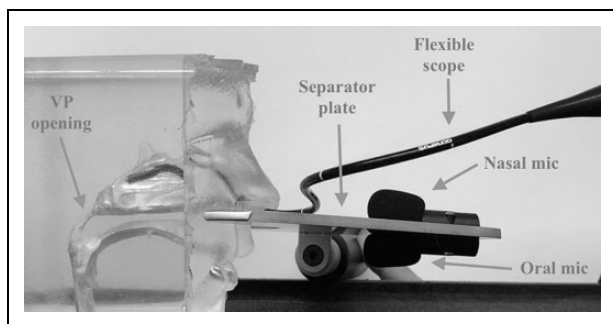


Figure 1. Illustration of the experimental setup. High-speed video (HSV) nasopharyngoscopy measurements were taken by connecting a flexible nasopharyngoscope to a high-speed video camera (not shown). The flexible nasopharyngoscope was inserted through the nose to the level of the nasopharynx to capture video of the velopharyngeal (VP) mechanism during speech. Acoustic data were captured with a customized system similar to a nasometer (see text for details).

difference in the image sampling rate. The MM1a and MM2a multimedia files show video clips from participants where the HSV images are subsampled in order to imitate the image sampling rate of standard nasopharyngoscopy. The nasal audio recording of each participant is added to these clips so that the reader can hear the nasal rustle. The other multimedia files (MM1b-d, MM2b) show the HSV data played back at 30 frames/second for the first /s/ sibilant sound each time the participant says “sixty.” The playback rate in these clips is much lower than the acquisition rate, which gives the perception of slow motion. In fact, it will take about 167 seconds to view 1 second of data that are captured at 5000 frames/second and played back at 30 frames/second. The files labeled MM1b, MM1c, and MM1d correspond to the participant shown in MM1a and the file labeled MM2b corresponds to the same participant as in MM2a. Bubbling of nasal secretions is clearly shown in M2b, while in M1b-d, the “bubbling” is more of a wavy displacement of the secretion. In our view, none of these clips show the velum vibrating (ie, velar flutter).

The audio recordings were done with similar hardware used in the nasometry procedure; 2 microphones (1/4", Type 4958, Brüel and Kjær, Duluth, Georgia) were connected to the top and the bottom of a separation plate. Windscreens were used to cover the microphones to prevent interference from airflow directed toward the microphones. The separation plate was made of aluminum (5 mm thickness) and its shape was based on the plate used with the Nasometer II system (PENTAX Medical, Montvale, New Jersey). The separation plate was placed against the child's upper lip to create an acoustic barrier between the oral and nasal audio recordings. This customized system for audio recording was needed in order to capture a wider range of frequencies than the Nasometer II system. The microphones used in the current study measure down to 15 Hz (at ± 3 dB) compared with a 50 Hz lower limit of the microphones that are used with the Nasometer II (KayPENTAX, 2010). The acoustic data from the microphones were captured at 51.2 kHz using a data acquisition system (NI 9234, National

Instruments, Austin, Texas) and customized LabVIEW code that digitized and converted the data to a “.wav” audio file. The initiation of the audio recording and imaging data (from the HSV nasopharyngoscopy) was synchronized using an input/output module (NI 9402; National Instruments).

Each patient performed specific speech tasks that included repetitive production of the pressure-sensitive syllable (/pa/), repetitive production of a word with plosive and fricative phonemes (“sixty”), and prolongation of a sibilant phoneme (/s/). The patients were asked to continuously repeat these speech samples, while the data were recorded (which lasted a little over 20 seconds). The order of producing these speech samples varied between patients, as did the total number of repetitions each patient was able to produce. The goal was to collect as many samples of pressure-sensitive consonants as possible for analysis. Therefore, a standard protocol of speech samples from each participant was not followed.

Data processing: Imaging data from HSV nasopharyngoscopy. The dynamic motion of the velum was quantified by analyzing the light intensity in each image (ie, frame) of the HSV data. The image sensor on the HSV camera is made of thousands of cells (ie, pixels) that convert light intensity into electrons. In the basic operation, the image is captured by reading the value in each cell (ie, accumulated charge), and this analog signal is then digitized. Typically, the light intensity that is captured by each pixel of the HSV camera changes according to the distance of the surfaces in the field of view: closer surfaces appear brighter in an image while surfaces that are further away appear darker. By tracking the increase or decrease in the total light intensity in each image over time (by summing the values from all the pixels in each image), one can determine whether a surface is moving toward or away from the camera. This image processing technique is commonly done with interferometry, which is used to analyze surface displacement and/or vibrations (Hariharan, 2010).

In the current study, the relative displacement of the velum and the pharyngeal walls can be quantified from the HSV data by tracking the change in the total light intensity from each image over time (Figure 2). An image of an open VP port (Figure 2A) would be overall characterized by darker regions that come from the cavity of the VP port. On the other hand, an image showing a closed VP port (Figure 2B) would be characterized with brighter regions because of the increased amount of light reflecting off the tissue. Consequently, the total value of the light intensity would be lower in the former image and higher in the latter. This process of summing the light intensity in each HSV image can thus capture gross dynamic motions at the superior aspect of the VP port, such as the elevation of the velum (Figure 2C).

In addition to the gross motions of the velum, summing the light intensity in the HSV images into an HSV intensity waveform can also capture the dynamics of finer motions (insert in Figure 2C). These smaller and rapid variations in the light intensity waveform can stem from surface vibrations. In the current study, these rapid variations occur when there is a

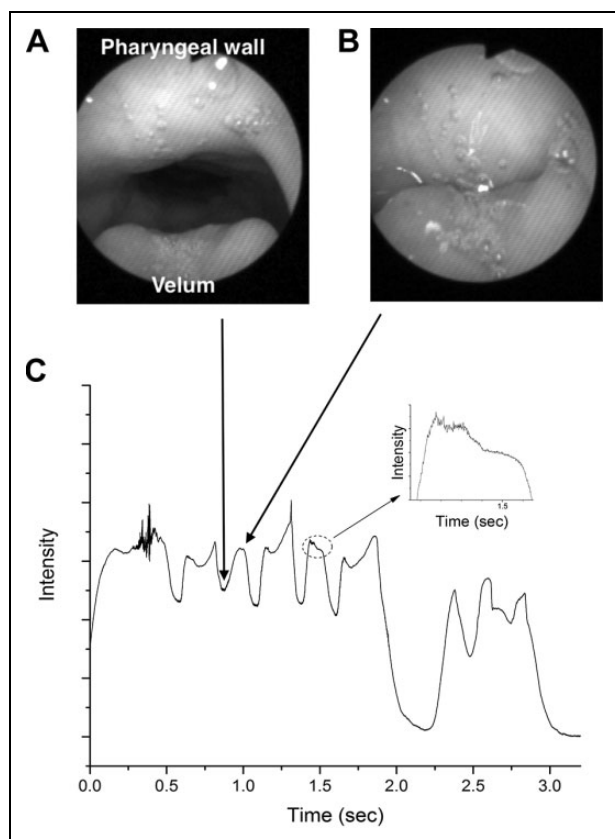


Figure 2. Illustrative images of (A) an open and (B) a closed velo-pharyngeal (VP) port. C, The temporal dynamics of the VP port is captured by summing the light intensity in the images of the high-speed video (HSV) data. This summation technique can capture the gross motion of the velum as well as (insert graph) fine periodic motions, such as bubbling or velum vibrations.

periodic process of the velum vibrating and/or secretion bubbling. The former can occur during voiced phonemes in normal speech with a closed VP valve (Dang et al., 2016). In the current study, the frequency information from the HSV intensity signal was extracted and correlated with the acoustic data to determine whether these periodic mechanisms generate sound.

Data processing: Acoustic data. After applying a low-pass filter to the acoustic data with a cutoff at the Nyquist frequency (the highest reliable frequency based upon the sampling rate), the duration of each pressure-sensitive phoneme (/s/ and /p/) was identified from the audio signal of each participant. The beginning and ending points of each phoneme were determined from the “frication noise” and/or the “gap” columns in the nasal spectrogram (Figure 3). In the illustrated example, the durations for 3 repetitions of the stop gap of /p/ are between 0.11-0.18, 0.33-0.41, and 0.56-0.68 seconds. Similarly, the duration for the sustained /s/ is between 1.18 and 1.68 seconds. These durations were then used to analyze and compare the signals from the nasal and oral microphones.

Spectral and Cross-Spectrum Analyses

Spectral and cross-spectrum analyses were performed on the acoustic waveforms (for both nasal and oral signals) and the HSV intensity waveform (Figure 4). For each phoneme, a fast Fourier transform (FFT) was calculated for the duration of the phoneme. Because the acquisitions of the audio and imaging data were synchronized, the timing information from the audio spectrograms (ie, beginning to end for each phoneme) was also used to segment the HSV intensity waveform. Thus, the length of each segment that was used for FFT analysis of the HSV data always matched the length of each segment from the acoustic data.

Bubbling frequency. The bubbling frequency on the superior aspect of the VP port was determined from the HSV intensity waveform (eg, Figure 2C). The rationale behind finding the frequency from the HSV data was to examine whether the bubbling process is periodic process and, if so, examine whether it generates sound that can be captured in the nasal acoustic signal. Spectral analysis for the bubbling frequency was done by taking the FFT of the HSV intensity waveform and applying a low-pass filter with a cutoff at the Nyquist frequency (2500 Hz; Figure 4A and B). The bubbling frequency was taken as the peak frequency in the spectrum; if no bubbling occurred during the phoneme, then the bubbling frequency was assigned a value of 0. The validation process for determining the frequency value was done by selecting one phoneme case at random from each participant and computing the frequency by manually determining the number of frames over which the displacement motion of the secretions occurred. This manual validation process was also performed for all cases where fewer than 4 bubbling cycles were observed during the pressure-sensitive phoneme.

Bubbling cycles per phoneme duration. The number of bubbling cycles during each phoneme was determined by counting the number of bubbles (or the wavy displacement of the secretion, c.f. M1b-d) that occurred during the time segment of each phoneme. It was expected that this number would vary depending on the duration of the phoneme and its location within a speech utterance (eg, beginning, middle, or end). Therefore, the number of bubbling cycles for each phoneme was divided by the duration of the phoneme to enable a more equal comparison across the different cases. The rationale for calculating this variable (bubbling cycles per phoneme duration) was to assess whether there is a minimum number of bubbles that needs to occur in order for this process to affect the acoustics.

Area between cross-spectrum curves. A cross-spectrum analysis was done to assess the correlation between the oral audio recordings versus the HSV signals and then between the nasal audio recordings versus the HSV signals (Figure 4C and D). Simply stated, the cross-spectrum analysis is a correlation analysis, but instead of using the waveforms, the cross-spectrum relates the frequency spectra of 2 signals to each other. This

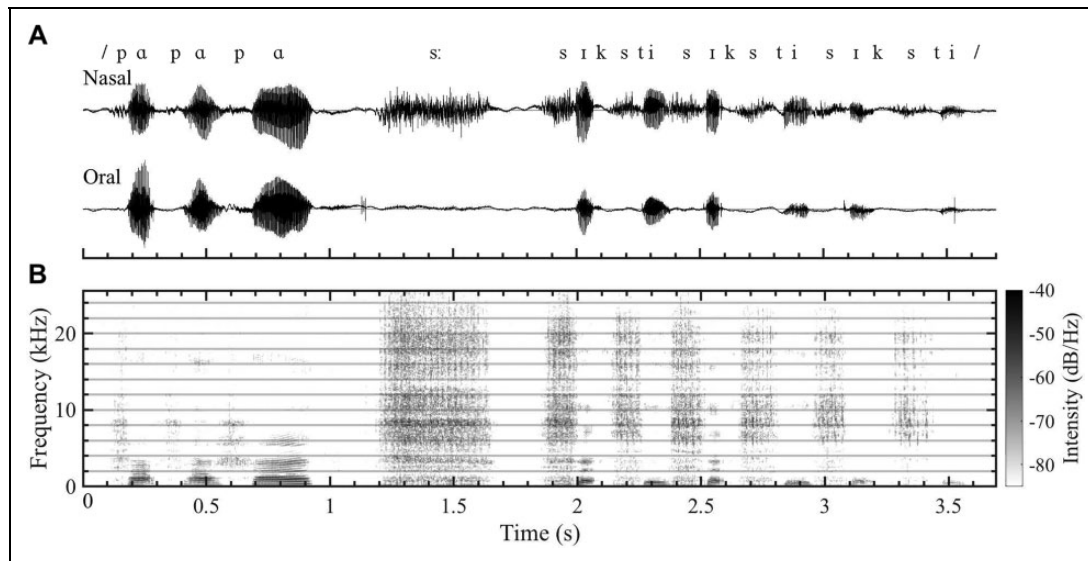


Figure 3. Illustrative acoustic data during fricative phonemes from a participant. A, Nasal (top) and oral (bottom) audio waveforms with phonetic annotation. B, Corresponding nasal spectrogram.

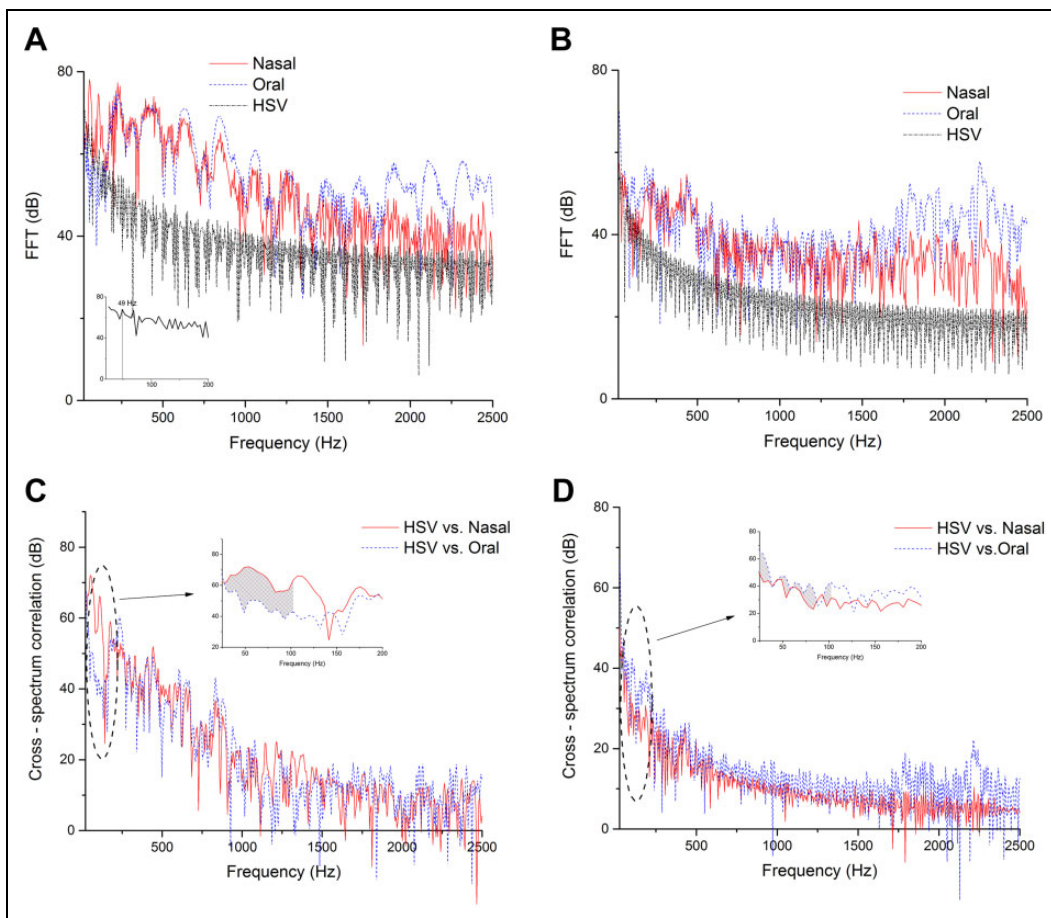


Figure 4. Spectral and cross-spectrum analysis computed from acoustic recordings and high-speed video (HSV; based on its light intensity waveform) in a single participant. A, Power spectra for /s/ phoneme with bubbling; insert graph shows a zoomed-in spectrum of the HSV signal in the 0 to 200 Hz range with peak at 49 Hz. B, Power spectra for /s/ phoneme without bubbling. C, Cross-spectrum analysis for /s/ phoneme with bubbling and (D) Cross-spectrum analysis for /s/ phoneme without bubbling. Inserts plots show close-ups of the 0 to 200 Hz range; shaded regions illustrate the area used to quantify the difference in area between cross-spectrum curves.

Table 1. Cases of Bubbling for All Phoneme.^a

Phoneme	/sss/	Beginning /s/	Middle /s/	/pa/	Row Total
No bubbling ($\pm$ Area)	1 (1/0)	27 (9/18)	28 (6/22)	36 (0/36)	92 (16/76)
Bubbling observed ($\pm$ Area)	22 (7/15)	47 (39/8)	46 (32/14)	28 (3/25)	143 (81/62)
Column total ($\pm$ Area)	23 (8/15)	74 (48/26)	74 (38/36)	64 (3/61)	235 (97/138)

^aThe “ $\pm$ Area” indicates the number of cases for which a positive/negative value was computed for the area between the cross-spectrum curves. See text for details.

analysis determines the relationship between 2 signals as a function of their frequency (Oppenheim and Schafer, 2014).

The rationale of correlating the audio recordings with the HSV images was to assess whether the periodicity of the secretion bubbling process can affect the acoustics in the nasal cavity. If a periodic mechanism (eg, bubbling and/or velar vibrations) occurs in the nasopharynx, it will likely be captured by the HSV and the nasal microphone signal, but not the oral microphone. It is therefore expected that the cross-spectrum analysis would show stronger correlation between the HSV and the nasal signals, especially around the bubbling frequencies (ie, 20-100 Hz).

To quantify the difference in “strength” of the cross-spectrums, the difference in area under the cross-spectrum curves (Mooers, 1973) was computed in the 20 to 100 Hz range (shaded regions in the insert plots of Figure 4C and D). This frequency band was selected to account for the highest bubbling frequency ($\sim$ 100 Hz) and the lower frequency limit of the microphones ($\sim$ 15 Hz). The area under the HSV versus oral cross-spectrum was subtracted from the area under the HSV versus nasal cross-spectrum. Thus, a positive value indicates that in the 20 to 100 Hz range, the cross-spectrum between the HSV and the nasal signals is stronger than with the cross-spectrum between HSV and the oral signal. A negative value for the area difference indicates that the opposite is true.

Results

Secretion bubbling was observed on the superior aspect of the VP port in all 10 participants. The bubbling was not continuous throughout the speech tasks in all participants. Instead, it occurred intermittently during the production of pressure-sensitive phonemes. The data from the 10 patients yielded a total of 235 pressure-sensitive phonemes, of which 143 (61%) cases were marked with bubbling (Table 1). Across all participants, bubbling was not observed during vowels.

When the participants repeated the word “sixty,” the number of bubbling cycles during the duration of the sibilant phoneme /s/ ranged from 0 to 21. Inter- and inpatient variations were also noted in durations of these sibilant phonemes (within a 50-350 milliseconds range). For the sustained sibilant /s/ and the stop gap /p/, the number of bubbling cycles ranged from 0

to 31 with 130 to 2160 milliseconds duration range, and 0 to 25 with 20 to 200 milliseconds duration range, respectively. No consistencies were observed in the transient time of the bubbling (ie, the time from the beginning of the phoneme until bubbling begun) nor in how long the bubbling process lasted within the phoneme duration. However, the process always ceased before or during the transition to the next speech sound.

The effect of the secretion bubbling on the acoustics was very clear (Figure 5). One of the main observations from the current study is that the bubbling during each phoneme case was a periodic process. The frequency of the bubbling in all cases occurred in the 18 to 100 Hz range, with the majority occurring between 20 and 40 Hz (Figure 5A). By plotting the bubbling frequency as a function of the number of bubbles that were observed for the duration of that sound, a random distribution is apparent, suggesting that the bubbling frequency is independent of the number of bubbles. In other words, the bubbling frequency is random but consistently occurs in the range of 18 to 100 Hz.

The majority of the cases where bubbling was not observed during the phoneme sound were marked by stronger cross-spectrum between the oral and HSV signals (ie, negative area). The data for the area between the cross-spectrum curves are plotted on different abscissas because of their range and dispersion (see caption for details). The area data seem to be independent of the bubbling frequency (Figure 5B) and somewhat dependent on the number of bubbles that were observed during the phoneme (Figure 5C).

An interesting observation can be made of cases where no bubbling occurred and the magnitude of HSV versus nasal cross-spectrum was higher (ie, zero frequency with positive area value). All of these cases occurred on the sibilant sounds and there was no evidence for a periodic mechanism in the nasal signal audio signal (ie, there were no dominant peaks in the 20-100 Hz range). Because there were no dominant frequencies in either the HSV or the audio signals, the cross-spectrum calculations are essentially correlating noise with noise. As explained in the “Methods” section, the cross-spectrum of the HSV will be stronger with the louder audio signal. It suggests that in these cases, the acoustic energy coming from the nasal cavity is stronger than the acoustic energy that is formed by the oral constriction.

Discussion

The purpose of the current study was to determine whether secretion bubbling is a mechanism that affects the sound generated in the nasal cavity with small VP opening. The results above confirm this hypothesis by showing that the frequency of the bubbling, which is measured by a visual technique, is also captured using audio measurement. This indicates that the bubbling affects the dynamics of the pressure fluctuations in the nasal cavity, which makes the acoustic field that is perceived as sound.

Better understanding is needed for the mechanism/s that turn nasal emission from inaudible to audible or to “severe and

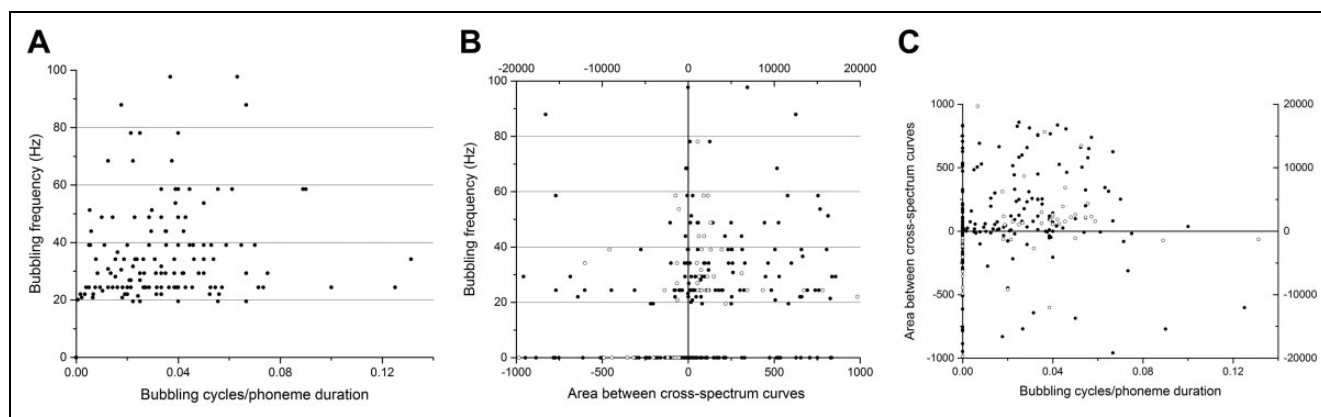


Figure 5. Bubbling versus acoustic data. A, Dominant frequency versus number of cycles of bubbling. B, Bubbling frequency as a function of the difference in area between the cross-spectrum curves. The closed and open symbols are shown on the bottom and top abscissas, respectively. C, Difference in area between the cross-spectrum curves as a function of bubbling cycles. The closed and open symbols are shown on the left and right ordinates, respectively.

distorting.” It has been assumed for decades that turbulent air-flow in the nasal cavity produces the loud and severely distorting form of audible nasal emission in cases of a small VP opening. Turbulent sound, however, is typically perceived as “noise” because its acoustic energy is spread over a wide (broadband) frequency range. It will therefore lack the periodic component that characterizes the severe form of nasal emission and is often perceived as “blowing a raspberry” or “air going through a sphincter.” The development of turbulent airflow in the nasal cavity is likely to be the source that turns nasal emission from inaudible (in cases of large VP opening) to audible (Sundström and Oren, 2019).

The current study shows that secretion bubbling above the VP opening is a periodic process that can affect the sound that is formed in the nasal cavity. Sundström and Oren (2019) showed how turbulent airflow is formed by the sudden expansion in space in the posterior nasal cavity. This turbulent air-flow interacts with the bubbling, which in turn modifies the turbulent sound. The interaction between the turbulent flow and the bubbling adds the periodic components to the turbulent sound. It changes the turbulent sound in the same manner that vibrating lips modulates the sound when one is blowing raspberries. The modification of the turbulent sound by the periodic bubbling mechanism changes the perception of the nasal emission from audible to loud and severe.

Zajac and Preisser (2016) argued that the modification of the turbulent sound by a periodic process was done by the vibrating margins of the velum. The vertical striations that were shown in the nasal spectrograms of their study, which indicate that some periodic process occurs simultaneously with the turbulence sound, are also shown in the acoustic measurements in the current study (see Figure 3). Unlike Zajac and Preisser (2016), the current study includes direct visualization of the VP opening while the acoustic data were captured, and no velar vibrations are observed. In our opinion, the secretions on the superior aspect of the velum were insufficient to act as a liquid damper and as such conceal low-amplitude tissue vibrations

that might occur underneath. Velar vibrations were sometimes observed during utterances of voiced consonants and vowels. These vibrations are likely caused by transvelar coupling of sounds (Dang et al., 2016). Spectral analysis of the HSV images showed that in all such cases, the velum was vibrating at the dominant frequency that was calculated from the oral microphone (ie, vocal fold oscillation frequency). On the other hand, bubbling was always observed when the vertical striations in the acoustic measurements occurred. Furthermore, the periodicity of these vertical striations matches with the bubbling frequency. These evidences strongly suggest that the modification to the turbulent sound that turns audible nasal emission into nasal rustle is due to bubbling of the secretions on the superior aspect of the VP opening.

The finding that sound is generated by the interaction of nasal airflow and the moving secretions is further supported by clinical observations that nasal rustle can be reduced if the nasal secretions are removed by suction or having the patient blow her/his nose (McWilliams et al., 1990). With no secretion present near the superior aspect of the VP port, there is no mechanism that would interact with the nasal airflow to generate the undesired sound that is perceived as nasal rustle. Turbulent airflow is still likely to form in the nasal cavity in the absence of secretion. However, the degree to which nasal emission is perceived as audible (eg, more audible, less audible, but not severe or distorting) will vary based on the specifics of the nasal geometry (Sundström and Oren, 2019).

Using the HSV nasopharyngoscopy offers some benefits over the standard technique for research purposes. In particular, HSV allows the examiner to better discern the velar function during specific phonemes (Hashemi Hosseinabad et al., 2019). For example, changes in velar motion (eg, degree of velar excursion, closing rate, etc) can be captured for different syllables during normal speech. This information can be used to improve models for velar function by also considering the phonetic inventory. In the current study, HSV nasopharyngoscopy also revealed that visual observations with standard

nasopharyngoscopy can sometimes be inaccurate in regard to bubbling. For example, in the clip that simulates a standard nasopharyngoscopy image acquisition rate (MM1a), one can perceive that there was bubbling. However, the “slow motion” clips (M1b-d) show wavy motion of the secretion instead of a complete bubble. The perception of bubbling might develop because standard nasopharyngoscopy undersamples the periodicity at which the wavy displacement motion of the secretion occurs. Correct modeling of the secretion motion is important for studies that seek to determine how bubbling modifies the turbulent sound.

The HSV can only act as support and additional imaging tool to the standard nasopharyngoscopy in a clinical setting. The time it takes to review the images and the lack of meaningful audio recording when the HSV data are played back in slow motion make it impractical to use HSV nasopharyngoscopy in clinics. The cost associated with high-speed cameras is another reason why this technique is typically only used in research.

Implications of this Study

One important observation from the current study is that secretion bubbling generates significant sound energy between 20 and 100 Hz—a range that is not analyzed by the nasometry systems typically used in clinics. These nasometry systems apply a 300 to 750 Hz band-pass filter to augment the hypernasality characteristics that are typically associated with larger VP openings (KayPENTAX, 2010). By filtering out the frequency range where the secretion bubbling occurs, the magnitude of the acoustic energy computed from the nasal microphone is lowered, thus lowering the overall value for the nasometry score. This filtering process can explain why, at times, patients who are perceived to have audible nasal emission caused by VPI have a low nasometry score value (similar to normal speech). These patients likely have a small opening where very audible secretion bubbling occurs; however, the effect on the nasometry score is reduced because of the filtering process used.

Understanding the mechanism that produces the severe and loud form of nasal emission is instrumental in clinical decision-making for patients with VPI as a result of craniofacial disorders. Clinicians can develop future therapies for treating speech disorders due to VPI by eliminating the source of the undesired sound in the nasal cavity. It is therefore important to know, in cases of small VP opening, whether that source is from the turbulent airflow (nasal turbulence), tissue vibrations (velar flutter), secretion bubbling (nasal rustle), or some combination of these.

The results of this study suggest that the interaction of the nasal airflow with nasal secretions is likely the main mechanism that produces the severe form of audible nasal emission. As such, even a small VP opening can be highly symptomatic and therefore require surgical intervention for permanent correction. A temporary fix for reducing the severity of the audible nasal emission would be management of secretions.

Study Limitations

The present study only examined data of specific pressure-sensitive phonemes that are assumed to maximize the incidences of bubbling. Future work will need to expand these speech samples and generalize these findings to other speech sounds. Speech samples with more syllables should especially be considered. The current study also did not include airflow measurements that could be used to infer information about the speech effort of each participant. Participants who were less receptive to the nasopharyngoscopy procedure seemed to increase their rate of speech and/or increase/decrease the loudness of their voice. These changes can directly affect the formation of secretion bubbling and airflow parameters (eg, oral pressure, flow rate, etc). Lastly, the effect of the flexible scope on the nasal acoustics, if any, also needs to be better understood. The presence of a scope might change resonance characteristics of the nasal cavity because of the change (reduction) in cavity volume compared to the cavity volume without the scope. Therefore, the undesired sound from the nasal cavity with versus without a nasopharyngoscope may be perceived differently by a listener. However, the sound generated by the bubbling formation is independent of the presence of a flexible scope; thus, the conclusions of this study are likely to hold.

Conclusions

Using simultaneous measurements of acoustics and HSV nasopharyngoscopy, the current study shows that secretion bubbling plays a significant role in the mechanism that generates the undesired sound when there is nasal emission and a small VP opening. The bubbling modifies the turbulent sound, making the audible nasal emission louder and more severe.

Future research is planned to further validate the hypothesis that the secretion bubbling acts as the mechanism that generates the sound identified as nasal turbulence/rustle. In a future study, audio recordings of speech containing nasal rustle will be filtered based on the frequency information from the HSV nasopharyngoscopy images. Clinicians will be asked to rate whether nasal rustle is present before and after the filtering process. In this way, it will be possible to determine whether the distortion of speech from audible nasal emission is reduced or eliminated when specific frequencies that relate to the bubbling are targeted.

Declaration of Conflicting Interests

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Supplemental Material

Supplemental material for this article is available online.

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